

Sable Arm Power Budget

Team Number:	303
Project Name:	Sable
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Version:	1

All Major Components	Component Name	Part Number	SupplyVoltageRange	#	AbsoluteMaximumCurrent (mA)	TotalCurrent(mA)	Unit
	9G RC Servo	SER0039	+0-5V	3	300	900	mA
	Stepper motor	SM-42HB34F08AB	+12V	1	1330	1330	mA
	Stepper motor driver	TMC260C-PA	+3-5.25V	1	12	12	mA
	LED	LTST-M670TBKT	+0-3.3V	5	20	100	mA
	ESP32 Microcontroller	ESP32-S3-WROOM-1	+2.7-3.6V	1	355	355	mA
+12V Power Rail	Component Name	Part Number	SupplyVoltageRange	#	AbsoluteMaximumCurrent (mA)	TotalCurrent(mA)	Unit
	Stepper motor	SM-42HB34F08AB	+12V	1	1330	1330	mA
					Subtotal	1330	mA
					Safety Margin	25%	
					Total Current Required on +12V Rail	1662.5	mA
c1. Regulator or Source Choice	+12V Battery		N/A	1	1662.5	1662.5	mA
					Total Remaining Current Available on +12V Rail	0	mA
+5V Power Rail	Component Name	Part Number	SupplyVoltageRange	#	AbsoluteMaximumCurrent (mA)	TotalCurrent(mA)	Unit
	9G RC Servo	SER0039	0-5V	3	300	900	mA
					Subtotal	900	mA
					Safety Margin	25%	
					Total Current Required on +5V Rail	1125	mA
c2. Regulator or Source Choice	+5V Buck Converter	AP1509-50SG-13	+4.5-22V	1	2000	2000	mA
					Total Remaining Current Available on +5V Rail	875	mA
+3.3V Power Rail	Component Name	Part Number	SupplyVoltageRange	#	AbsoluteMaximumCurrent (mA)	TotalCurrent(mA)	Unit
	Stepper motor driver	TMC260C-PA	+3-5.25V	1	12	12	mA
	ESP32 Microcontroller	ESP32-S3-WROOM-1	+2.7-3.6V	1	355	355	mA
	LED	LTST-M670TBKT	+0-3.3V	5	20	100	mA
					Subtotal	467	mA
					Safety Margin	25%	
					Total Current Required on +3.3V Rail	583.75	mA
c3. Regulator or Source Choice	+3.3V Buck Converter	LM2575-3.3WU-TR	+4-40V	1	1000	1000	mA
					Total Remaining Current Available on +3.3V Rail	416.25	mA
External Power Source 1		Component Name	Part Number	SupplyVoltageRange	Output Voltage	AbsoluteMaximumCurrent (mA)	TotalCurrent(mA)
Power Source 1 Selection		+12V Battery	N/A	N/A	+12V	5000	5000
							mA
Power Rails Connected to External Power Source 1		+12V Power Rail	N/A	N/A	+12V	1662.5	1662.5
		+5V Buck Converter	AP1509-50SG-13	+4.5-22V	+5V	2000	2000
		+3.3V Buck converter	LM2575-3.3WU-TR	+4-40V	+3.3V	1000	1000
					Total Remaining Current Available on External Power Source 1	337.5	mA
External Power Source 2		Component Name	Part Number	SupplyVoltageRange	Output Voltage	AbsoluteMaximumCurrent (mA)	TotalCurrent(mA)
Power Source 2 Selection		+9V 5A Wall Power Supply		90-264 VAC	9V	5000	5000
							mA
Power Rails Connected to External Power Source 2		+12V Power Rail	N/A	N/A	+12V	1662.5	1662.5
		+5V Buck Converter	AP1509-50SG-13	+4.5-22V	+5V	2000	2000
		+3.3V Buck converter	LM2575-3.3WU-TR	+4-40V	+3.3V	1000	1000
					Total Remaining Current Available on External Power Source 2	337.5	mA
		Component Name	Part Number	SupplyVoltageRange	Capacity(mAh)	RequiredByRegulators	
		Battery	(full part number)	+12V	8000	4662.5	
					Battery Life	1.715817694	hours